STACKS, DERIVED CATEGORIES OF SPACES: UNBOUNDED-RELATIVE-MAYER-VIETORIS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1978-1993 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 1995-2035, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 10 of queue_spaceperfect.json (pb=1526, stmt 1978-1993, proof 1995-2035).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. gate_ok=true: raw-interval guard false positive (public-items.jsonl upstream-frame coords); precard_gate.py shift-corrected check cleared this label 2026-10-14 (gate_sp01.sh).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-unbounded-relative-mayer-vietoris' also exists in stacks-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0152: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2035–2035. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 014Z: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2035–2035. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 079P: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2035–2035. Cited by the retained statement or proof but

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GW: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1889–1909. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08FI: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2035–2035. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $(U \subset X, V \to X)$ be an elementary distinguished square. Denote $a = f|_U : U \to Y$, $b = f|_V : V \to Y$, and $c = f|_{U \times_X V} : U \times_X V \to Y$ the restrictions. For every object E of $D(\mathcal{O}_X)$ there exists a distinguished triangle*

$$Rf_*E \to Ra_*(E|_U) \oplus Rb_*(E|_V) \to Rc_*(E|_{U \times_X V}) \to Rf_*E[1]$$

in $D(\mathcal{O}_Y)$. This triangle is functorial in E.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing E. We may assume $\mathcal{I}^n$ is an injective object of $Mod(\mathcal{O}_X)$ for all n, see Injectives, Theorem 079P. Then Rf_*E is computed by $f_*\mathcal{I}^\bullet$. Similarly for U, V, and $U \cap V$ by Cohomology on Sites, Lemma 08FI. Hence the distinguished triangle of the lemma is the distinguished triangle associated (by Derived Categories, Section 014Z and especially Lemma 0152) to the short exact sequence of complexes

$$0 \to f_*\mathcal{I}^\bullet \to a_*\mathcal{I}^\bullet|_U \oplus b_*\mathcal{I}^\bullet|_V \to c_*\mathcal{I}^\bullet|_{U \times_X V} \to 0.$$

To see this is a short exact sequence of complexes we argue as follows. Pick an injective object $\mathcal{I}$ of $Mod(\mathcal{O}_X)$. Apply f_* to the short exact sequence

$$0 \to \mathcal{I} \to j_{U,*}\mathcal{I}|_U \oplus j_{V,*}\mathcal{I}|_V \to j_{U \times_X V,*}\mathcal{I}|_{U \times_X V} \to 0$$

of Lemma 08GW and use that $R^1 f_*\mathcal{I} = 0$ to get a short exact sequence

$$0 \to f_*\mathcal{I} \to f_*j_{U,*}\mathcal{I}|_U \oplus f_*j_{V,*}\mathcal{I}|_V \to f_*j_{U \times_X V,*}\mathcal{I}|_{U \times_X V} \to 0$$

The proof is finished by observing that $a_* = f_*j_{U,*}$ and similarly for b_* and c_*. $\square$